INFS3701 — Twelve-Month Cost Estimate

Corporate Cloud Storage Gateway · Region: **US East (N. Virginia) us-east-1**

Built for the group deliverable using **AWS Pricing Calculator** (https://calculator.aws/) line items and current On-Demand public pricing. Figures below are a planning estimate (ex-tax); re-open the calculator with the same assumptions to refresh the share link before submission.

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1. Solution being priced (optimised lab architecture)

Estimate matches the deployed architecture: two private **t3.micro** app servers behind an internet-facing **Application Load Balancer**, **Multi-AZ RDS PostgreSQL db.t3.micro** (20 GiB), private encrypted **S3**, **Secrets Manager**, CloudWatch dashboard/alarms, and SNS email. Egress uses a **single NAT Gateway** (cost optimisation vs dual-NAT). VPC, S3 gateway endpoint, and security groups have no material hourly fee.

2. Key assumptions

• Region: us-east-1 only; pricing model: On-Demand; 730 hours/month × 12 months.
• Traffic: low academic/demo load (~light ALB LCU ≈ 1; ~10 GB/month NAT data processing; ~5 GB S3; ~20 GB outbound negligible after free tier caution — outbound included lightly via ALB/LCU assumptions).
• Compute: 2 × t3.micro Linux; 8 GiB gp3 EBS each; no EC2 detailed monitoring.
• Database: PostgreSQL Multi-AZ db.t3.micro; 20 GiB gp3; minimal backup retention; no Performance Insights.
• Networking: 1 NAT Gateway + 1 associated public IPv4; ALB across 2 AZs with public IPv4 charges modelled as 2 additional addresses.
• Ops: 3 Secrets Manager secrets; 1 CloudWatch dashboard; ~1 GB/month log ingest; SNS email for alarms.
• Not included: Route 53 domains, ACM public cert fees (lab uses imported cert), data transfer beyond light demo use, Reserved/Savings Plans.
• Learner Lab: use small sizes, capture evidence, then tear down — this 12-month estimate is the production-style course deliverable, not the short lab bill.

3. Monthly cost breakdown (USD)

Service / component	Monthly (USD)
NAT Gateway (1×)	33.30
Public IPv4 (3×: NAT+ALB)	10.95
Application Load Balancer	22.27
EC2 t3.micro ×2	15.18
EBS gp3 8GiB ×2	1.28
RDS PostgreSQL db.t3.micro Multi-AZ	24.82
RDS storage gp3 20GB Multi-AZ	4.60
S3 Standard 5GB + light requests	0.61
Secrets Manager (3 secrets)	1.20
CloudWatch (dashboard + 1GB logs)	3.53
SNS email (minimal)	0.50
Total per month	**118.24**
Total for 12 months	**1,418.93**

4. Optimisation choices (how cost was kept down)

• **One NAT Gateway** instead of two (~halves the largest networking line; disclose AZ-A egress dependency in the design doc).
• **t3.micro / db.t3.micro** and 8–20 GiB disks — enough to prove architecture, not production throughput.
• **S3 gateway endpoint** keeps app↔S3 traffic off NAT.
• Short lab sessions: deploy → evidence/video → **delete stack**; do not leave NAT/ALB/RDS running between sessions.
• Dual-NAT alternative would add roughly another **~$33–37/month** (second NAT + IPv4), i.e. ~**$400+/year** more, without changing the marking core.

5. Twelve-month summary

Estimated steady-state cost in us-east-1: **$118.24 per month**, approximately **$1,418.93 over 12 months** (ex-tax). Rebuild/confirm these lines in AWS Pricing Calculator before submission and attach a calculator screenshot or share URL in

the appendix if required by your tutor.

Sources: AWS Pricing Calculator (calculator.aws); Amazon EC2, VPC, ELB, RDS, S3, Secrets Manager, CloudWatch, SNS public pricing for us-east-1. Estimate only — actual fees vary with usage.